**PASIG CATHOLIC COLLEGE**

**College Department**

**BACHELOR OF SCIENCE IN INFORMATION TECHNOLOY**

**IT107 - CASE STUDY**

# "School Equipment Borrowing and Inventory Management System for Ugong Pasig National High School"

**Team Member**

**Belen, Laiza Mae V.**

**Borce, Iyanne Coline C.**

# TABLE OF CONTENTS

I. INTRODUCTION

II. COMPANY PROFILE/ CLIENT BACKGROUND

III. Stakeholders of the System

IV. System Features / Program Functions

V. Technology Used for development

VI. Program Output

# I. INTRODUCTION

The School Equipment Borrowing and Inventory Management System for Ugong Pasig National High School (UPNHS) is a web-based information system designed to make school equipment inventory and borrowing transactions more organized and manageable. The system provides a convenient way to manage equipment information, monitor available quantities, process borrowing requests, and maintain borrowing records in one interface. It aims to help the school keep track of its equipment and provide a more organized process for both the administrator and users when borrowing and returning school property.

The system has two main types of users: the administrator and regular users. The administrator is responsible for managing the equipment inventory, including adding, editing, deleting, searching, and filtering equipment records. The administrator can also review borrowing requests, approve or reject them, and confirm the return of borrowed items after checking their condition. Meanwhile, users can view available equipment, submit borrowing requests, provide the purpose of borrowing, and monitor the status of their requests. Once a request is approved, the system records the borrowing date and sets a return deadline of three days after approval. It also allows users to report returned equipment, while the administrator records whether the item is returned in good condition, damaged, or lost. These features help keep borrowing transactions and equipment records updated and organized.

The system is developed using HTML, CSS, and JavaScript, with JSON for data organization and browser local storage for saving information. HTML provides the structure of the system, including its login page, equipment inventory, borrowing forms, and records tables. CSS is used to design the interface and organize its layout for easier navigation. JavaScript handles the system's main functions, such as account sign-in, equipment management, request processing, return confirmation, quantity updates, and record monitoring. JSON organizes the equipment details, user accounts, and borrowing transactions into a structured format, while local storage keeps the saved information available even after the browser page is refreshed. The system also includes a JSON export feature that allows the administrator to download a backup of the stored data.

This project is important because proper equipment management helps schools monitor their resources and keep track of items borrowed by students and other authorized users. When equipment information and borrowing transactions are difficult to monitor, checking availability, identifying current borrowers, and tracking return deadlines can become challenging. The proposed system addresses these concerns by organizing equipment details and borrowing records in one place, making information easier to access and manage. It also helps the administrator monitor pending requests, borrowed items, overdue equipment, and the condition of returned items. Overall, the project applies the concepts of web development and data management learned in IT107 while providing a practical way to support the organization and monitoring of school equipment at UPNHS.

## II. COMPANY PROFILE/ CLIENT BACKGROUND

Ugong Pasig National High School (UPNHS) is a public secondary school that provides education to Junior High School and Senior High School students. The school uses different types of equipment and learning resources to support classroom instruction, laboratory activities, school programs, and other educational purposes. These resources include projectors, laboratory materials, speakers, extension cords, remote controls, and library books.

Managing school equipment requires proper recording of available quantities, equipment conditions, borrowers, and return schedules. When equipment information and borrowing transactions are handled through manual records, monitoring availability and tracking borrowing activities can become difficult, especially when several transactions take place.

The proposed School Equipment Borrowing and Inventory Management System is designed to provide a digital way to organize equipment inventory and borrowing requests. It allows the administrator to maintain equipment records, review and process borrowing requests, monitor borrowing periods, and record equipment returns. Users can sign in, view available equipment, submit borrowing requests, and report returned equipment for confirmation. The system serves as a management tool for school personnel responsible for equipment and provides a more organized process for authorized school users who need to borrow school resources.

## III. STAKEHOLDERS OF THE SYSTEM

| Stakeholder | Role / Description | Interest in the System |
|---|---|---|
| School Principal | The person responsible for approving equipment borrowing requests. | To monitor and approve requests and ensure proper use of school equipment. |
| Admin Office / Equipment Staff | Users who manage borrowing requests, equipment records, and borrower information. | To keep equipment records organized and updated. |
| Students / Teachers / School Personnel | Authorized users who request and borrow school equipment for schoolrelated purposes. | To check equipment availability and submit borrowing requests easily. |

## IV. SYSTEM FEATURES/PROGRAM FUNCTIONS

| Module | Function Description |
|---|---|
| 1. Login Module | Allows the administrator and users to sign in using their login credentials. It also allows new users to create an account. |
| 2. Equipment Display Module | Displays available equipment, including its name, available quantity, and condition, for users to view before submitting a borrowing request. |
| 3. Equipment Inventory Module | Allows the administrator to add, edit, and delete equipment records, including equipment names, quantities, and conditions. |
| 4. Search and Filter Module | Allows the administrator to search for equipment by name, filter items by availability status, and sort equipment by the highest or lowest available quantity. |
| 5. Borrowing Request Module | Allows users to select available equipment, enter their grade and section or department, provide the purpose of borrowing, and submit a borrowing request. |
| 6. Request Approval Module | Allows the administrator to review, approve, or reject pending borrowing requests after checking equipment availability. |
| 7. Borrowing Period Module | Automatically records the borrowing date and sets the return deadline to three days after the administrator approves the request. |

| | |
|---|---|
| 8. Return Module | Allows users to report returned equipment and enables the administrator to confirm the return and record the equipment condition as Good, Damaged, or Lost. |
| 9. Inventory Update Module | Automatically decreases the available quantity when a request is approved and increases it when the administrator confirms that the returned equipment is in good condition. |
| 10. Borrowing Records Module | Displays borrowing information, including the borrower, equipment borrowed, borrowing date, return deadline, actual return date, and current status. |
| 11. Dashboard and Summary Module | Displays the total number of equipment types, available quantity, pending requests, borrowed items, overdue items, returned items, damaged items, and lost items. |
| 12. Data Storage Module | Saves equipment records, borrowing requests, and user account information in browser local storage so that the saved data remains available after refreshing the page. |
| 13. Validation Module | Checks required information in borrowing requests and validates equipment details when the administrator adds or edits inventory records. It also checks equipment availability and prevents duplicate active requests for the same equipment by the same user. |
| 14. Notification Module | Displays alerts for successful account creation, request approval or rejection, and return record updates, as well as confirmation messages for important actions. |
| 15. Overdue Monitoring Module | Checks borrowing records and changes the status of borrowed equipment to Overdue when its return deadline has passed and the overdue-checking function runs. |
| 16. Request Cancellation Module | Allows users to cancel their own borrowing requests while the requests are still pending. |
| 17. Data Export Module | Allows the administrator to download a JSON backup containing equipment records, borrowing requests, and user account information. |

## V. TECHNOLOGY USED FOR DEVELOPMENT:

| Technology | Purpose |
|---|---|
| 1. HTML | Provides the structure of the system, including the login page, forms, tables, buttons, and dashboard sections. |
| 2. CSS | Controls the system's appearance, including its layout, colors, spacing, and responsive design. |
| 3. JavaScript | Handles the system's main functions, including login, equipment management, search and filtering, borrowing requests, request approval, return processing, and record updates. |
| 4. JSON | Represents equipment records, user accounts, and borrowing requests in a structured data format. It is used to convert data for storage and retrieval and to export backup files. |
| 5. Browser Local Storage | Stores equipment records, borrowing requests, and user account information in the browser so the data remains available after the page is refreshed. |

## VI. Program Output




**Figure 1. Login Interface**

The login page allows the administrator and registered users to access their respective accounts by entering their login credentials. It also provides an option for new users to create an account.

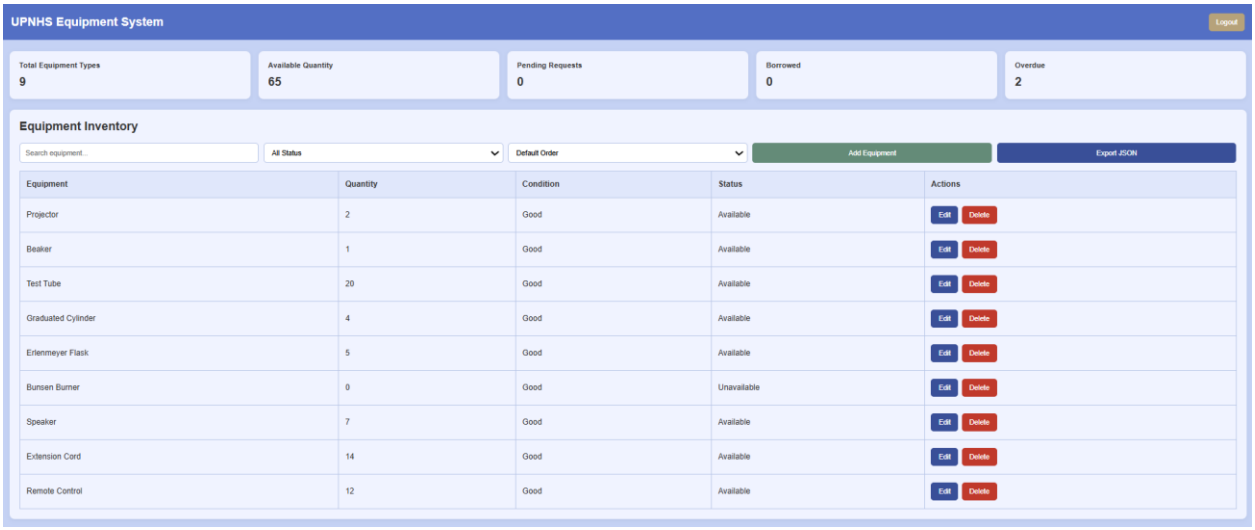



**Figure 2. Administrator Dashboard**

The administrator dashboard provides an overview of the school's equipment inventory and borrowing status. It displays the total number of equipment types, available quantity, pending requests, borrowed items, and overdue items. The Equipment Inventory section allows the administrator to search, filter, and sort equipment records, add new equipment, edit or delete existing records, and export data in JSON format. This interface helps the administrator monitor equipment availability and manage inventory records efficiently.



Borrowing Requests

| Borrower | Grade / Department | Equipment | Purpose | Request Date | Actions |
|---|---|---|---|---|---|

Borrowing Records

| Borrower | Equipment | Borrowed Date | Return Deadline | Actual Return | Status | Actions |
|---|---|---|---|---|---|---|
| Iyanne Coline | Projector | 2026-10-09 | 2026-10-12 | 2026-10-09 | Returned | - |
| Laiza Mae | Extension Cord | 2026-10-09 | 2026-10-12 | 2026-10-09 | Damaged | - |
| Kenzo Basty | Projector | 2026-10-09 | 2026-10-12 | 2026-10-09 | Damaged | - |
| Iyanne Coline | Test Tube | 2026-10-09 | 2026-10-12 | 2026-10-09 | Returned | - |
| Laiza Mae | Beaker | 2026-10-10 | 2026-10-13 | 2026-10-10 | Returned | - |
| Laiza Mae | Projector | 2026-10-10 | 2026-10-13 | - | Overdue | Confirm Return |
| Sarah | Beaker | 2026-10-10 | 2026-10-13 | - | Overdue | Confirm Return |

System Summary

Returned: 3
Damaged: 2
Lost: 0

**Figure 3. Borrowing Requests, Borrowing Records, and System Summary**

The administrator dashboard includes sections for managing borrowing requests, monitoring borrowing records, and viewing the system summary. The Borrowing Requests section displays the borrower's name, grade or department, requested equipment, purpose, and request date, along with available actions for processing requests. The Borrowing Records section presents the borrowing date, return deadline, actual return date, and current status of each transaction. The administrator can also confirm the return of overdue equipment and update its condition. The System Summary displays the total number of returned, damaged, and lost items, helping the administrator monitor equipment returns and maintain organized borrowing records.




**Figure 4. User Dashboard and Equipment Borrowing Form**

The user dashboard displays the available school equipment, including equipment names, available quantities, and conditions. The Borrow Equipment section allows users to select equipment, enter their grade and section or department, specify the purpose of borrowing, and submit a borrowing request for approval. The My Requests section enables users to monitor their submitted requests, including request dates, return deadlines, actual return dates, and current statuses. These features help users check equipment availability and track their borrowing transactions in one interface.

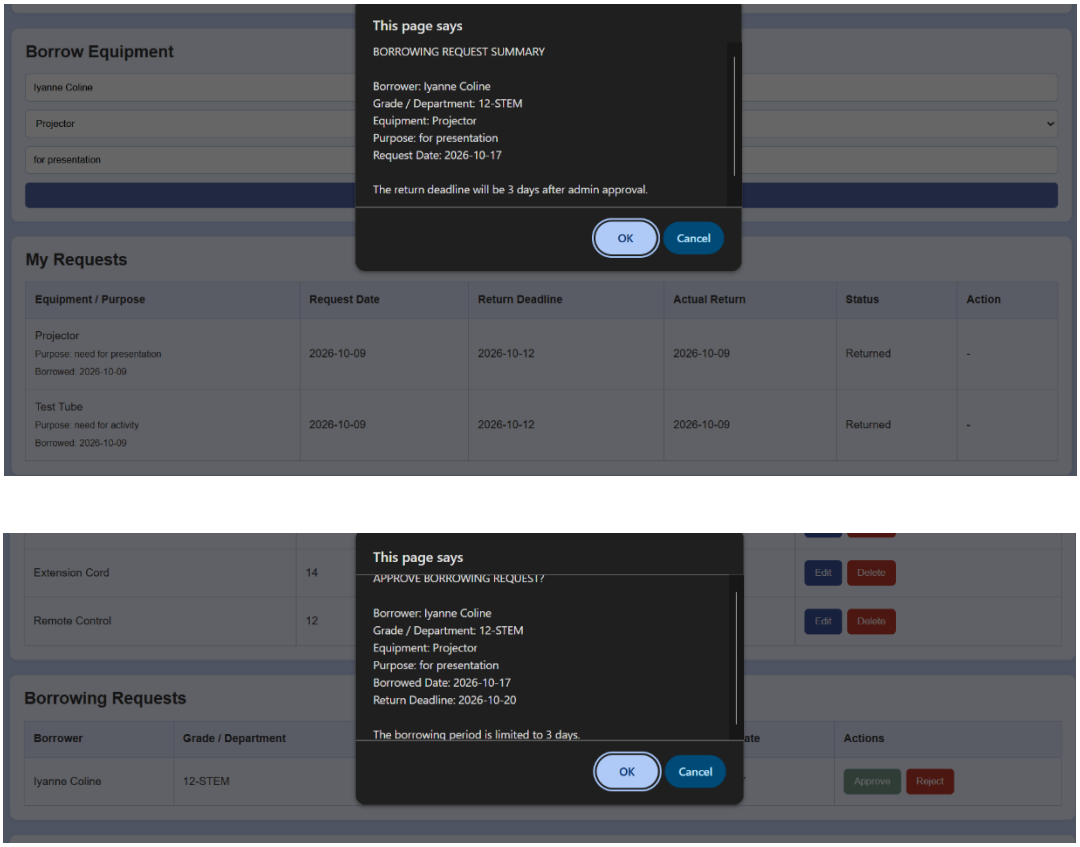



**Figure 5. Borrowing Request Submission and Approval Process**

The borrowing process begins when the user selects available equipment, enters the required information, and submits a borrowing request. Before submission, the system displays a request summary containing the borrower's name, grade or department, selected equipment, purpose, and request date for confirmation. Once submitted, the request appears in the administrator's Borrowing Requests section for review. The administrator can approve or reject the request. Upon approval, the system records the borrowing date, updates the available equipment quantity, and sets a return deadline three days after approval. This process helps ensure that borrowing transactions are properly recorded and monitored.

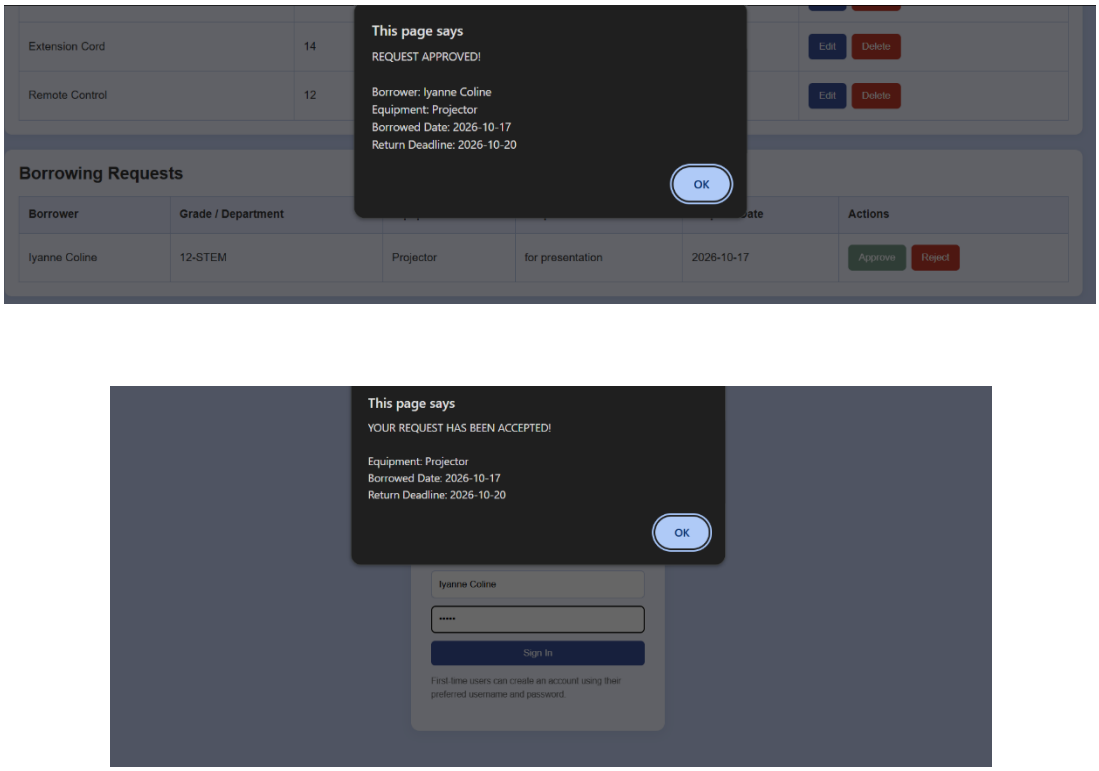



**Figure 6. Borrowing Request Confirmation and Approval Notifications**

The system displays confirmation messages to inform users and the administrator about the result of the borrowing request. Once the administrator approves a request, the user receives an acceptance notification, while the administrator receives an approval confirmation. These messages include the equipment name, borrowing date, and return deadline. The system sets the return deadline three days after approval, helping both parties keep track of the borrowing period.

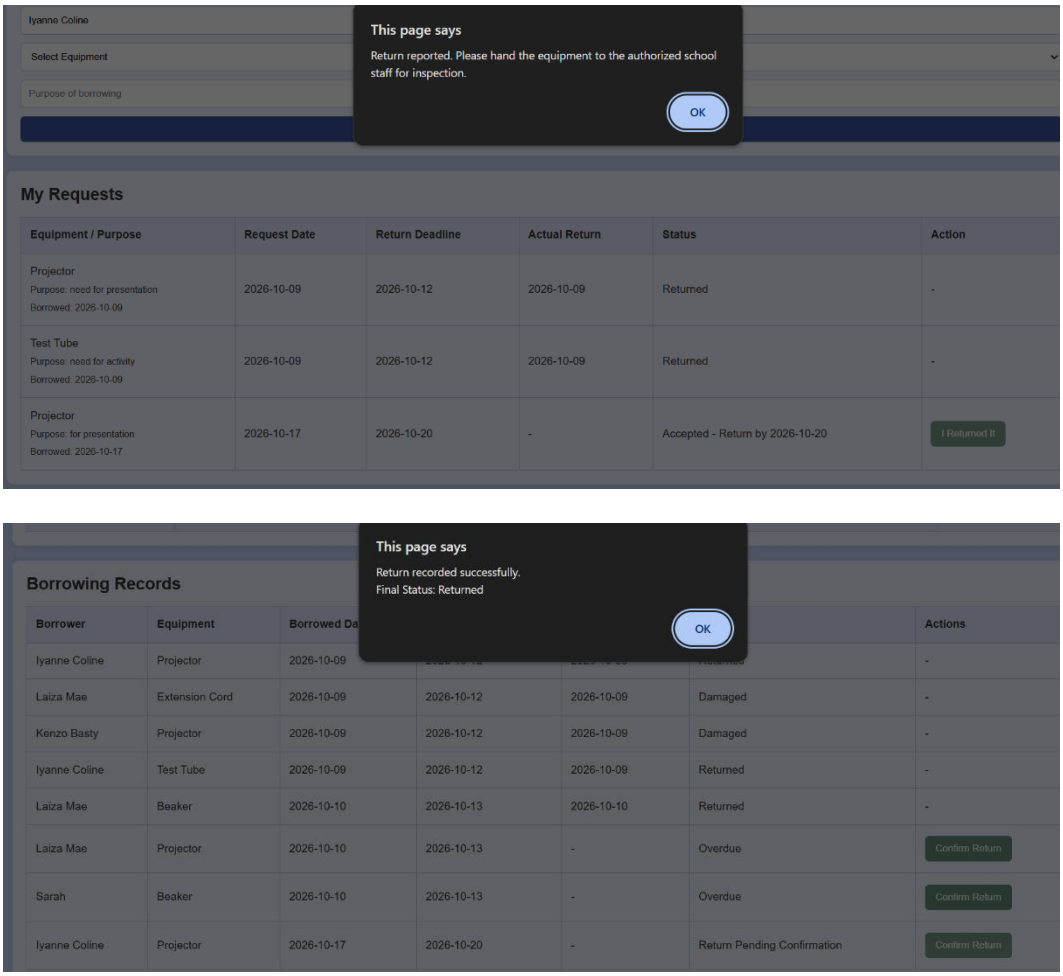



**Figure 7. Equipment Return Reporting and Administrator Confirmation**

The return process begins when the user clicks the “I Returned It” button to report that the borrowed equipment has been returned. The system displays a message reminding the user to hand over the item to authorized school personnel for inspection. Afterward, the administrator clicks “Confirm Return” to record the item's final condition as Good, Damaged, or Lost. Once confirmed, the system updates the borrowing record with the actual return date and final status. The available equipment quantity is also updated if the item is confirmed to be in good condition.

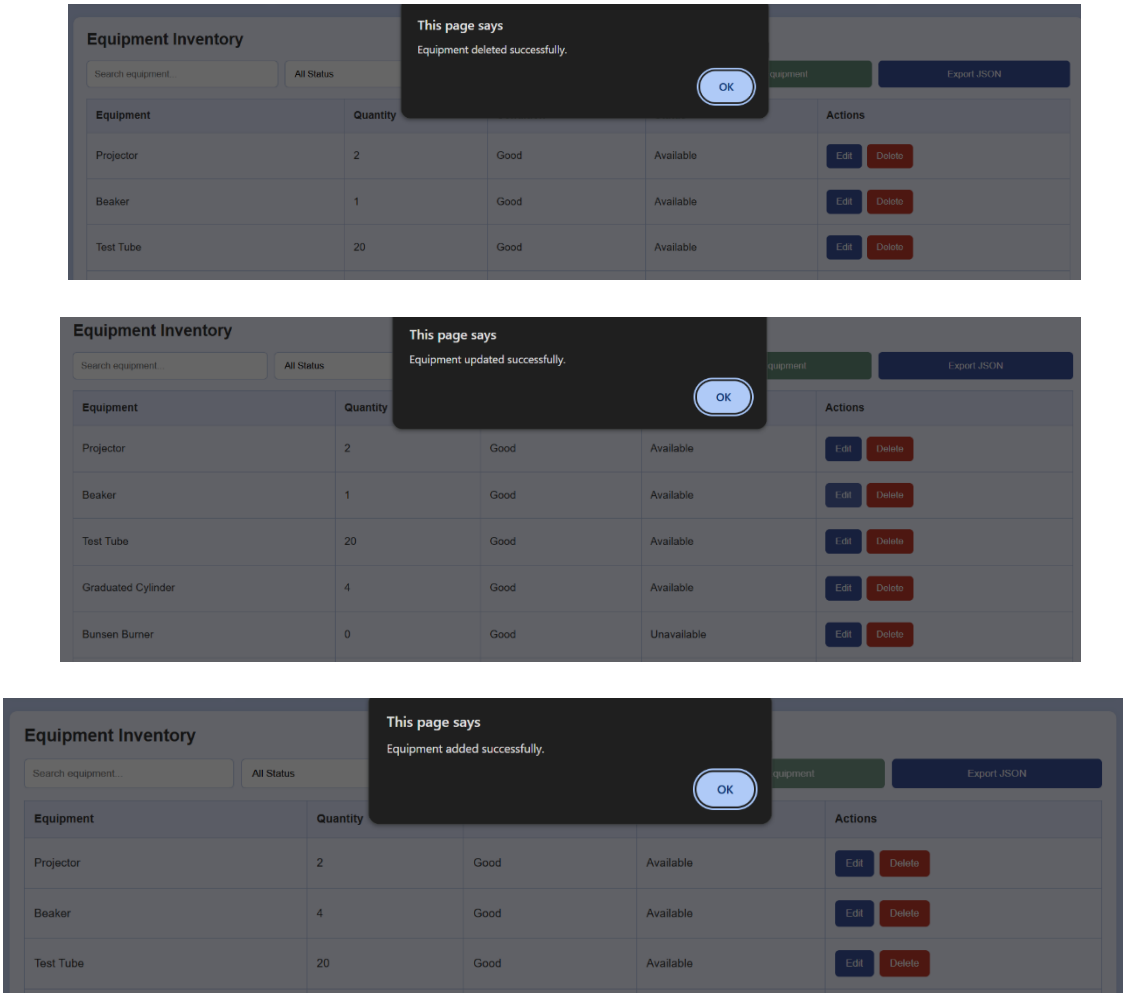



**Figure 8. Equipment Inventory Management Functions**

The equipment inventory management functions allow the administrator to add new equipment, edit existing equipment details, and delete equipment records when necessary. When adding or editing an item, the administrator can update its information, including the equipment name, quantity, and condition. These functions help maintain accurate inventory records, monitor equipment availability, and keep the system's information updated.

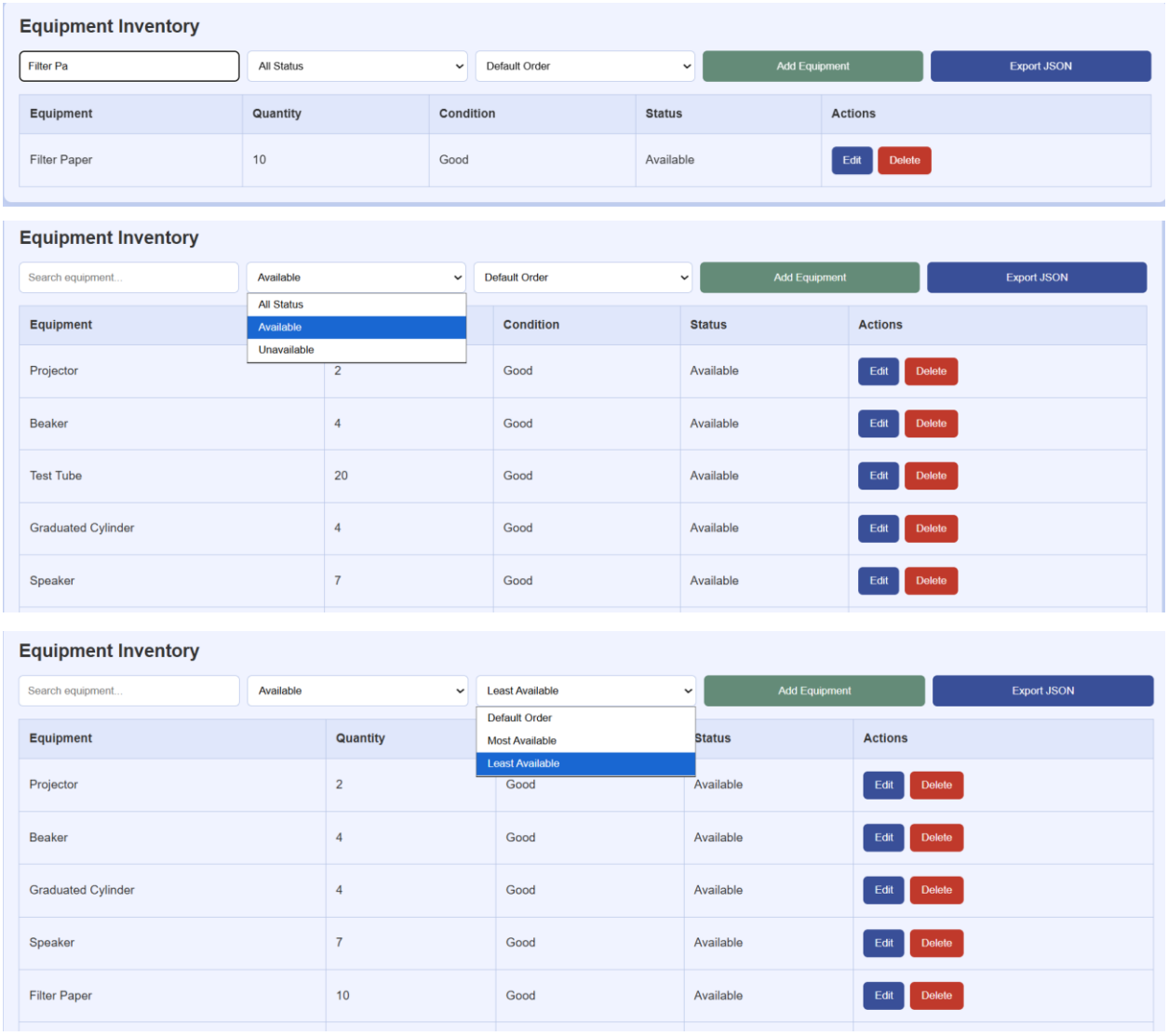
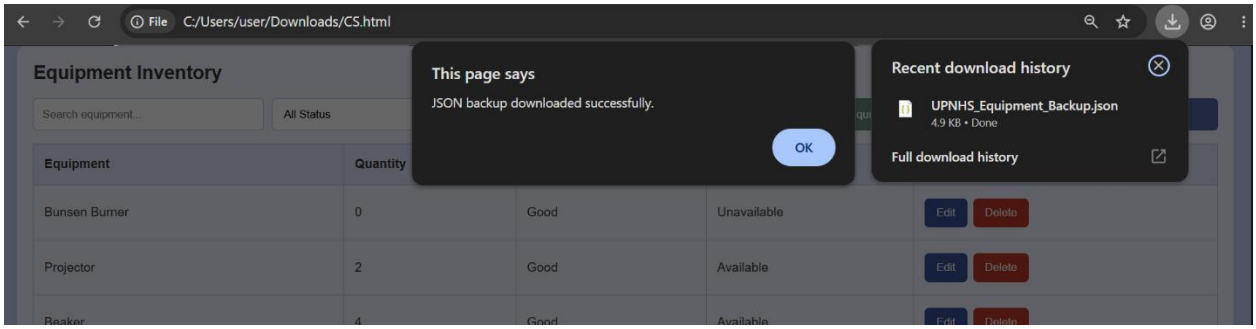



**Figure 9. Equipment Search, Filtering, and Sorting**

The equipment inventory provides search, filtering, and sorting functions to help the administrator find and organize equipment records. The search bar allows the administrator to locate specific equipment by name, while the status filter can display all equipment, available items, or unavailable items. The sorting option arranges the equipment list according to the default order, from most available to least available. These functions make it easier to check equipment availability and manage inventory records efficiently.



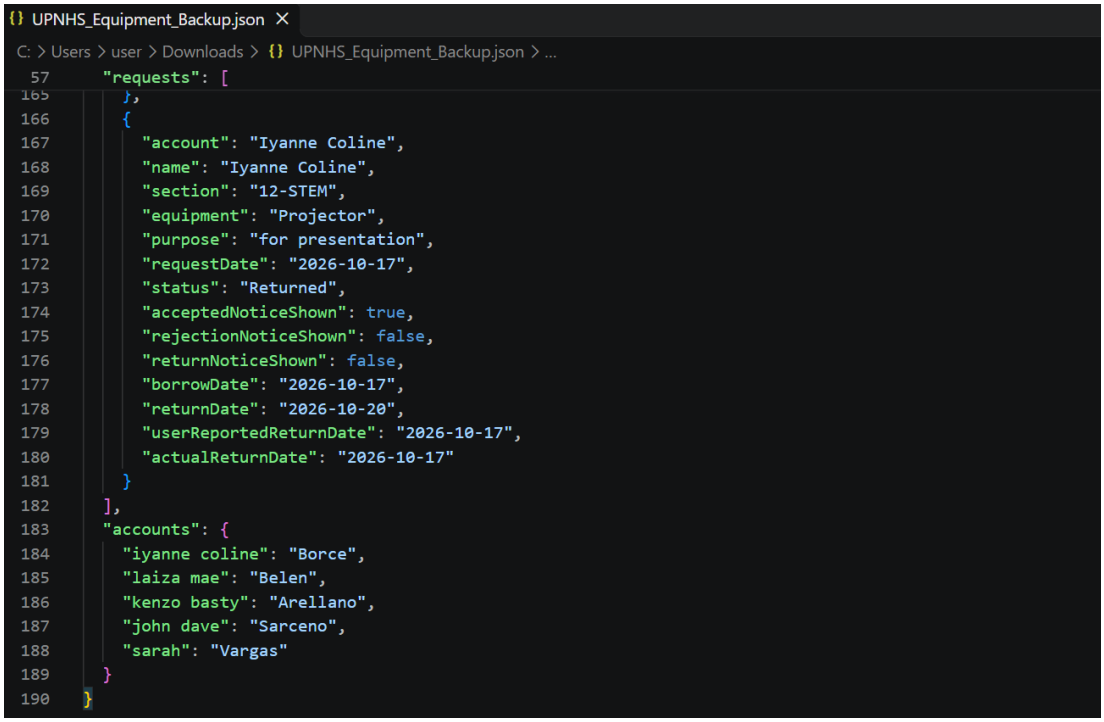




**Figure 10. JSON Data Export and Backup**

The system provides a JSON export feature that allows the administrator to download a backup of the stored equipment inventory, borrowing requests, and user account information. The downloaded file contains the records in a structured JSON format, including equipment details, borrower information, request dates, borrowing status, and return dates. The exported data can be opened and viewed in Visual Studio Code, making it easier to inspect the saved records. This feature helps maintain a separate copy of the system's data for backup and reference purposes.